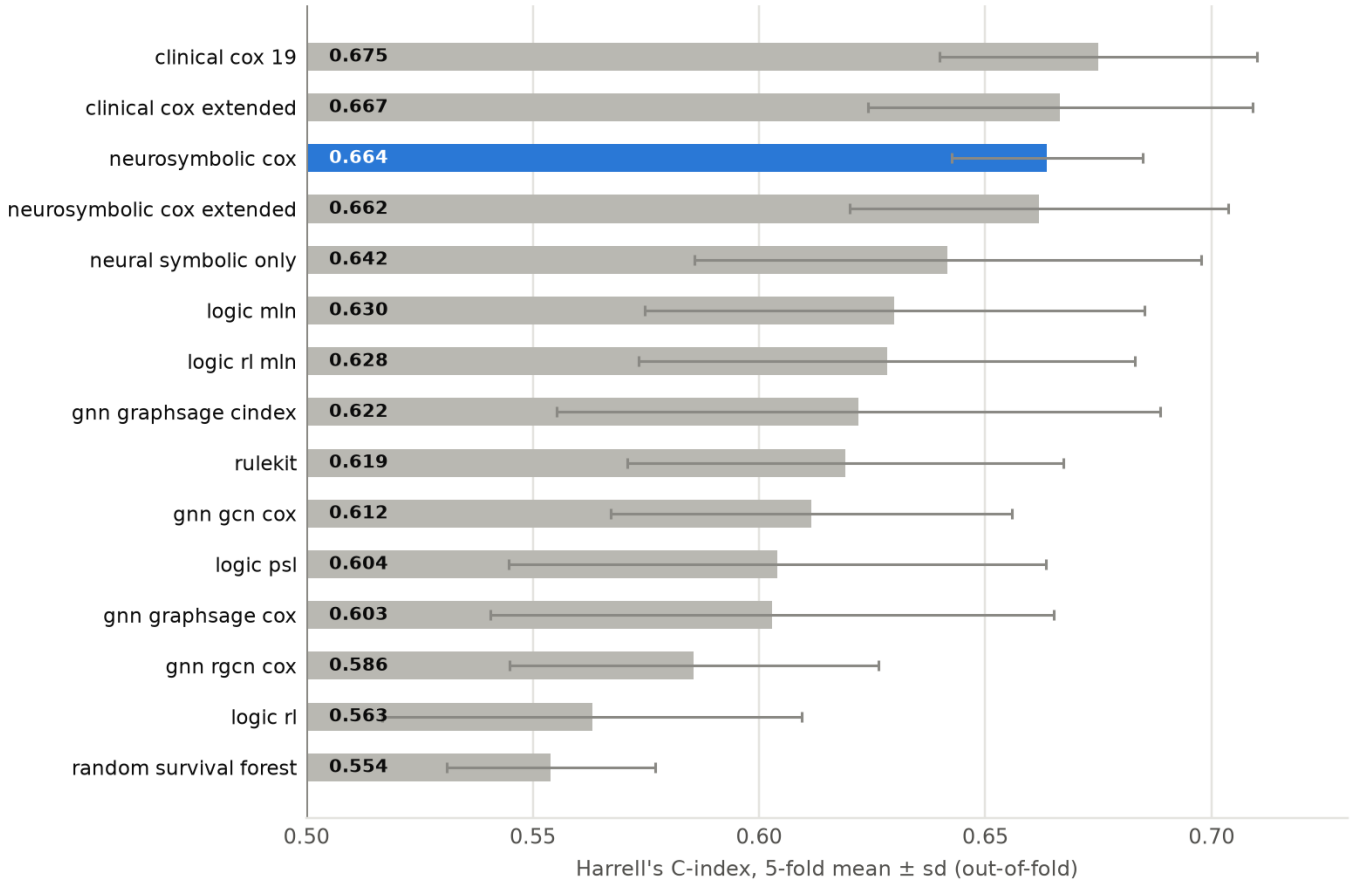
Cohort Overview: Neuro-Symbolic Progression Model

PrimeKG Patient Reasoning · Precision Oncology Research

1083 patients

Report date 31 Oct 2026

Model comparison (all predictions out-of-fold, same five folds)



Model	Harrell C	Uno C (120 mo)	AUC 12 mo	AUC 36 mo	AUC 60 mo	IBS 120 mo
clinical cox 19	0.669	0.616	0.691	0.710	0.621	0.134
clinical cox extended	0.666	0.657	0.687	0.677	0.633	0.135
neurosymbolic cox	0.660	0.616	0.673	0.702	0.613	0.134
neurosymbolic cox extended	0.663	0.655	0.681	0.678	0.631	0.135
neural symbolic only	0.617	0.606	0.614	0.638	0.605	0.139
logic mln	0.620	0.617	0.621	0.634	0.592	—
logic rl mln	0.612	0.615	0.618	0.617	0.575	—
gnn graphsage cindex	0.599	0.573	0.598	0.619	0.618	—
rulekit	0.592	0.623	0.544	0.610	0.597	—
gnn gcn cox	0.589	0.549	0.637	0.630	0.565	—
logic psl	0.604	0.571	0.628	0.598	0.574	—
gnn graphsage cox	0.595	0.556	0.585	0.637	0.589	—
gnn rgcn cox	0.564	0.590	0.506	0.578	0.567	—
logic rl	0.561	0.570	0.578	0.569	0.540	—
random survival forest	0.565	0.559	0.554	0.581	0.529	0.145

Lower integrated Brier score (IBS) is better; it is reported only for models that produce a survival curve.

Symbolic reasoners compared

Reasoner	Harrell C (mean ± sd)	Uno C	Formulas / fold	KG formulas
Survival MLN (Cox-MLN)	0.630 ± 0.055	0.617	32.8	63%
RL-structured survival MLN (engine)	0.628 ± 0.055	0.615	35.2	69%
RuleKit LR-Rules (baseline)	0.619 ± 0.048	0.623	7.8	10%
Survival PSL (collective)	0.604 ± 0.059	0.571	40.0	62%
RL rules (policy gradient)	0.563 ± 0.046	0.570	25.6	91%

The engine is chosen with the one-standard-error rule: among reasoners within one standard error of the best fold-mean C-index, the one with the richest reasoning is preferred (RL-structured MLN, then MLN, PSL, RL rules, RuleKit).

Predicting the month of progression

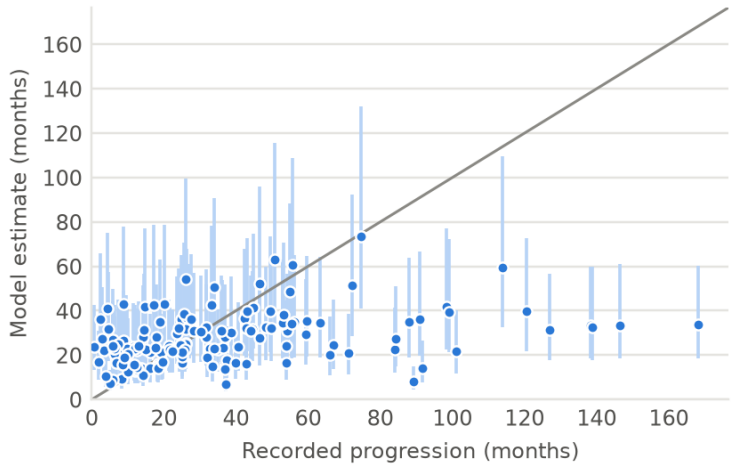
Estimator (held-out progressors)	MAE	Median |error|	Within 12 mo	Window accuracy	IQR coverage
conditional median tau60	23.2	18.2	33%	18%	37%
conditional median tau120	27.9	24.6	23%	39%	41%
conditional median tau180	59.2	62.0	7%	39%	33%
training progressor median	22.9	16.0	37%	19%	49%
lognormal aft on progressors	20.6	12.9	48%	34%	54%
loglogistic aft on progressors	20.1	12.9	48%	33%	57%

The bold estimator is the report's "model-estimated progression time". The training-progressor median is a constant baseline that ignores the patient.

Report groups

Group	Recorded PFS	Patients	Median model risk pct	Model window agreement
Group A · Early progression	≤ 12 months	32	77	12%
Group B · Progression in year 2	12–24 months	30	81	60%
Group C · Progression in year 3	24–36 months	27	71	37%
Group D · Late progression	> 36 months	56	51	29%
Group N · No progression recorded	censored	938	48	—

Model-estimated against recorded progression month (held-out progressors)



Each dot is one of the 145 progressors; the vertical bar is the model's interquartile range and the diagonal is perfect agreement. Estimates are compressed towards 15–45 months: early progressions are over-estimated and progressions after five years are under-estimated.